



Exploring the group holiday decision-making process with the support of technology



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ABSTRACT

Numerous studies have examined the decision-making process and the associated information search behaviour of tourists. However, very few have focused on how tourists make plans for their trips when travelling in groups. This paper adds to the knowledge of the decision-making process of tourist groups with a specific focus on intra-group interaction and the support of technology. This research adopted a mixed-methods approach, combining two studies. Study 1 employed a focus group to gain an understanding of the decision-making process within a tourist group, while Study 2 employed a questionnaire survey to explore how the use of technology in such a process varies among different categories of groups. Study 1 proposed a model for a group holiday decision-making process with four components: information acquisition, intra-group information sharing, knowledge transfer, and decision making. It was adapted from the “Infogineering Model” and the encoding and decoding theory in communication studies. Study 2 identified new influential factors on the use of technology by tourist groups for decision making. Ultimately, this research produced a number of theoretical implications, as well as practical implications in terms of how practitioners can better support Chinese tourists, family travellers, and tourists with little prior travel experience, and how technology can facilitate group holiday decision making.

1. Introduction

The expenditure on international tourism around the world has been following an increasing trend overall for the past two decades, apart from some small fluctuations, and expenditure by Chinese tourists abroad has been the highest in the world since 2012 (World Bank, 2017). Holidays play major roles in people's lives, providing opportunities for them to experience something different from their everyday routines (Decrop & Snelders, 2004). The process of trip planning is considered an important part of the whole travel experience. To understand the planning experience of tourist trips, existing studies have examined the decision-making process (Mathieson & Wall, 1982; Schmoll, 1977), and the factors that influence the decision-making behaviour of tourists (Engel & Blackwell, 1982; Gnoth, 1997; Goodall, 1991).

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In many cases, people tend to travel with a group of people, so that they may socialise, enjoy the company of each other, and better fit into their communities (Decrop, 2005). Existing studies have focused mainly on the distribution of roles within a household with regard to decision making about travel plans (Fodness, 1992; Wang, Hsieh, Yeh, & Tsai, 2004), and on the influence of children on family travel arrangements (Thornton, Shaw, & Williams, 1997; Carr, 2006). Meanwhile, other studies have investigated group cohesiveness and congruence within travel groups, including family and friend groups (Decrop, 2005).

Web 1.0, Web 2.0, and social media enable Internet users from all over the world to interact, communicate, and share ideas, thoughts, experiences, information, and foster relationships (Kaplan & Haenlein, 2010). Tourists, especially those who travel in groups, now rely on the Internet to obtain tourism information, to share information among themselves, and to make decisions. Current travel planning applications provide some functions for tourist groups to work on their itineraries together, such as Tripomatic (tripomatic.com); others enable discussions among group members, such as Planapple (planapple.com); while others make travel recommendations for groups, such as Pixmeaway (pixmeaway.com).

The existing research focuses mainly on how individual tourists make holiday decisions (Lew & McKercher, 2006; Mansfeld, 1993; Mathieson & Wall, 1982; Schmoll, 1977), and there is also a very limited number of studies that have explored group holiday decision making (Decrop, 2005). Furthermore, among these existing studies, many have focused on very specific characteristics of the group holiday decision-making process (Fodness, 1992; Carr, 2006; Decrop, 2005). A systematic investigation is required to provide a good understanding of how tourist groups make holiday decisions with the support of technology. Given that people group together based on different social structures, and exhibit significant differences in behaviour, the challenge for the researchers and practitioners is to be aware of such differences and to respond adequately (Prebensen, 2005). In addition, in this digital era, tourists in groups now use technologies to assist their group decision making. Very few studies have explored how tourists acquire and share information within groups with the support of technologies during the group holiday decision-making process. This research attempts to fill this research gap. It makes potential theoretical contributions by constructing a model which researchers may employ to understand how tourist groups obtain and communicate information with the support of technology to make holiday plans. It also has practical implications for practitioners in the tourism industry who may want to provide customised products, services, and technologies to decrease the gap between pre-purchase motives and activities pursued on holiday, and ultimately thereby enhance the overall decision-making experience of tourists travelling in groups (Ross & Iso-Ahola, 1991). The aims of this research are:

- To understand the group holiday decision-making process, supported by technology, with a specific focus on information acquisition and intra-group interaction (Study 1).
- To investigate how such a process, when supported by the use of technology, varies among different categories of tourist groups (Study 2).

This research employed methodological triangulation (Jupp, 2006) to investigate the group holiday decision-making process. It involved applying both group interviews and surveys to collect data in empirical studies. Group interviews were used to extract in-depth data about how tourists in groups make holiday decisions with the support of technology. Surveys were applied to acquire data about how such a process would be influenced by factors that are related to group characteristics. Triangulation is useful in the understanding and constructing of user behaviour because it provides complementary findings (Seybold, 2001). Qualitative interview data was suitable for gathering insights on participants' attitudes and thoughts about the group trip planning process, and any factors influencing this process, while surveys are appropriate for validating factors qualitatively among large populations.

2. Study 1

2.1. Aim of study 1

Study 1 collected data from interviews with 12 groups of tourists, to gain an understanding of the information acquisition process and the interactions between group members making decisions about travel plans, based on real-life scenarios.

2.2. Related work

2.2.1. Tourism decision-making process models

Existing works have studied tourism decision-making processes and have proposed implicit or explicit process models to describe such decision making (Lew & McKercher, 2006; Mansfeld, 1993; Mathieson & Wall, 1982; Schmoll, 1977). The common element in these models is that they deal with tourism in terms of space or time (Smallman & Moore, 2010).

Sequential models of holiday decision-making frameworks consist of several procedural stages to help understand the tourism decision-making process. Schmoll (1977) proposes a model with four main stages: motivation, information search, evaluation of alternatives, and decision making. Mathieson and Wall (1982) propose a tourism decision-making framework with the following steps: desire to travel, information collection and evaluation, travel decisions, preparation and experience, and evaluation of satisfaction. Spatial models for holiday decision making depict the spatial movement patterns of tourists within a destination (Lew & McKercher, 2006).

There are also some models which deal with the notion of tourists' decision making as emergent (that is socially constructed). The model proposed by Litvin, Goldsmith, and Pan (2008) describes the influence of eWOM (electronic word-of-mouth) in the holiday decision-making process, and discusses how technological and ethical issues can be harnessed by the emerging eWOM technologies.

Middleton and Clarke (2001) use a stimulus-response model to analyse tourist decision making from the aspects of needs, wants, goals, perceptions, and attitude. This research focuses on investigating the components of the group holiday decision-making process, and exploring the connections among such components.

2.2.2. Use of technology for tourism decision making

In this digital world, technologies allow tourists to search for and share tourism information on a larger scale. Existing studies have examined the use of technology (containing Web 1.0, Web 2.0 and social media) to assist people in organising holidays (Dipietro, Wang, Rompf, & Severt, 2007; Lo, Mckercher, Lo, & Cheung, 2011; Marlow & Dabbish, 2014; Sigala, 2010). Other studies have also investigated the actual behaviour of technology usage (e.g., Leung, Law, van Hoof, & Buhalis, 2013; FreshMinds Research, 2011). FreshMinds Research (2011) reports that, firstly, at the information gathering stage, people tend to obtain large amounts of information through browsing websites about hotels and attractions from social media websites such as Facebook, YouTube, and TravelBlog. Secondly, at the booking stage, people use travel comparison websites, such as Tripadvisor, Expedia, and Turisti per Caso, to acquire specific information and make reservations. Finally, after arriving at a destination, tourists tend to check fan-pages and feeds from social media or recommended websites, such as Lonelyplanet and Tripadvisor, to ensure their choices are in their best interests.

The aforementioned technologies have different characteristics in terms of purpose, role, and the domains of their application and some of these characteristics have already been explored in detail. Existing literature covers the need of tourists to search for different types of information, such as accommodation, entertainment, recreation, etc. (Dipietro et al., 2007), the use of different platforms, such as social network sites, online photograph albums, personal blogs, etc. (Lo et al., 2011), preferences for reading information in different formats, such as texts, slideshows, videos, etc. (Marlow & Dabbish, 2014), and the use of different types of features (services, functionalities, and interactivities) supported by technologies, such as geo-location based services (Sigala, 2010) and image-based electronic word-of-mouth (Bronner & Hoog, 2011).

2.2.3. “Infogineering Model” and encoding and decoding theory

In the group trip planning process, communication and interactions among group members are essential to make the group trip plans (Decrop, 2005). However, existing literature lacks a specific focus on how group members acquire and share information among themselves for the purposes of decision making. To address this limitation, this research adapted the “Infogineering Model” and the encoding and decoding theory to describe the process of group decision making.

As shown in Fig. 1, the “Infogineering Model” (IM) describes the connections among data, information, knowledge, and decisions. Decisions (informed actions) are made based on an individual's knowledge (an individual's understanding of the world), where such knowledge comes from information (captured useful data) or data (facts). On a surface level, decisions are made based on the information at hand. However, what has been adapted from the IM in this study is that information needs to be processed, interpreted and transferred into knowledge for further decision making. Here, the process for realising the transition between information and knowledge is defined as the encoding and decoding process in communication studies.

Hall (2001) proposes the encoding and decoding theory (EDT) in communication studies. The main idea is to transmit knowledge through a medium (discourse). As shown in Fig. 2, the senders encode the knowledge based on their own meaning structures and yield a “meaningful” discourse (message) to allow transmission. The receivers decode the message based on their own meaning structures (which may be different from the meaning structures of the senders). It is the knowledge that “has an effect”, influences, instructs, or persuades, with very complex perceptual, cognitive, emotional, or behavioural consequences. Misunderstanding arises when the meaning structures differ between senders and receivers. To illustrate the information acquisition and intra-group interactions for group trip planning, this research aimed to build a model, based on the IM and EDT.

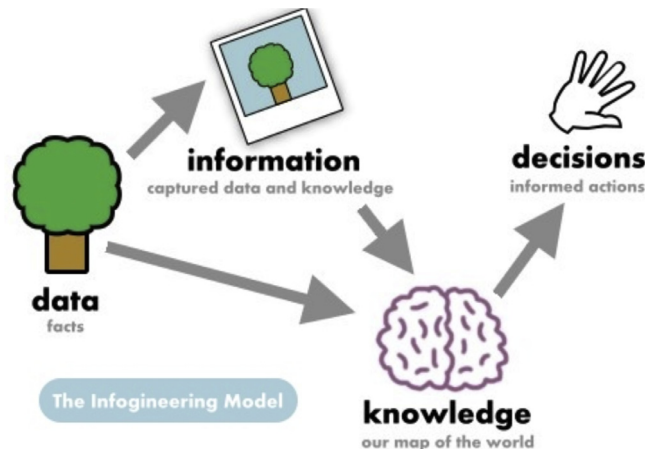


Fig. 1. The Infogineering Model (2017).

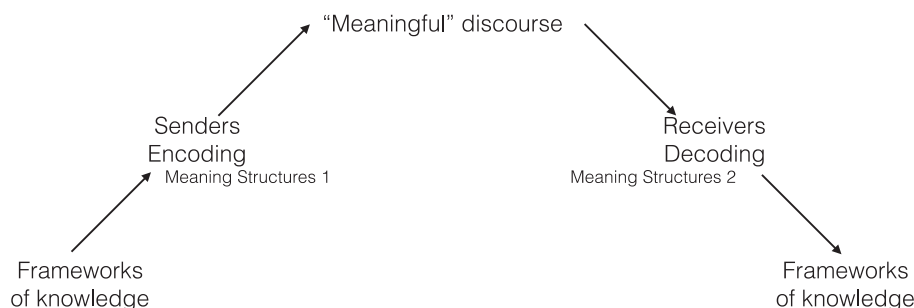


Fig. 2. Encoding and decoding theory (Hall, 2001).

2.3. Method

2.3.1. Study design

For the focus group interviews, Study 1 recruited holidaymakers who had travelled with a group of people in the previous three months. The interviews aimed to find both the facts and meanings of stories underlying the participants' experiences (Kvale & Brinkmann, 2009; McNamara, 2009). The questions for the focus group interviews were designed based on the IM and EDT (see details in Appendix A).

2.3.2. Participants

12 groups of participants took part in Study 1, following posts on social media and emails circulated around the University of Nottingham, which has campuses both in the UK and China. Each group had two or more members who had been to a holiday destination with which they were unfamiliar. The only requirement was that the participants had been travelling with at least one other person within the preceding three months. Participants consisted of family groups (3), friend groups (3), colleague groups (3), and mixed type groups (3). Appendix B lists the participants' profiles, including demographic information and the use of online media to make trip plans (measured using a 5-point Likert scale). In total, 24 participants (10 males, 14 females) took part in the interviews (the people highlighted in Appendix B did not take part in the study, but their involvement in the trip planning process was described by their group members who did take part in the study). To motivate potential participants to take part in the study, each group member was compensated for their time.

2.3.3. Data collection and procedure

Study 1 began with a brief introduction of the study and what was expected from the participants. After participants were recruited and had signed the consent forms, interview appointments were made for immediately after they had returned from their trips, when their memories were still fresh. The interviews were audio-recorded, and fully transcribed. At the end of the interviews, a short questionnaire was issued to collect the demographics and detailed travel information.

2.3.4. Data analysis

The interview data was qualitative in nature. Emergent Themes Analysis (Wong & Blanford, 2002; Sun, Sharples, & Makri, 2011) was employed to interpret the qualitative data, combined with the use of *a priori* coding (Sandelowski, 2000). The analysis began with a comprehensive read-through of the interview transcripts for all 12 groups, following which the transcripts were reviewed for coding purposes. Before beginning the emergent coding process, *a priori* codes were derived from existing work to provide the researchers with prior knowledge of the subject matter (Blair, 2015). The four characteristics of online tourism information (i.e., content, platform, format, and features) were the *a priori* codes employed to guide the emergent coding. Ultimately, the underlying connections and relationships were coded to answer the research questions.

2.4. Results and discussion

Based on our empirically grounded study, we constructed a model adapted from the IM and EDT. This model (see Fig. 3) illustrates the decision-making process among members of a group of tourists, with a specific focus on how tourism information is obtained, shared, and used within such a group.

2.4.1. Information acquisition

In the proposed model (Fig. 3), members in tourist groups obtain tourism information through two approaches: internal information acquisition (obtained internally from group members) and external information acquisition (obtained from the Internet, or through asking family members, friends, and colleagues). Participants explained that they searched through large amounts of information on the Internet, and they also searched different types of online tourism information. As shown in Fig. 3, four categories of online information and their detailed sub-categories have been identified.

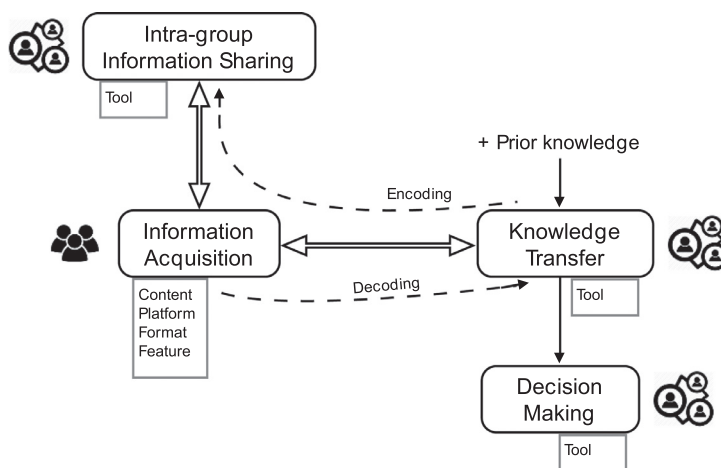


Fig. 3. The decision-making model for group holiday planning.

Firstly, participants searched different types of information *content*. This finding is very similar to Jafari's (1982) work on the essential elements of tourism when tourists choose tourism products and services. Secondly, participants searched different types of *platform* for information. Xiang and Gretzel (2010) put forward five categories of online social media for tourism information, based on Google search results. Thirdly, participants preferred different types of information *format*, such as texts, pictures, videos, panoramas, blogs, and review comments (Chen et al., 2014; Cox, Burgess, Sellitto, & Bultjens, 2009; Marlow & Dabbish, 2014). Finally, participants explained that they preferred different *features* of information, such as built-in maps, and recommendations based on geo-locations, etc. Sigala (2010) examines how customers perceive geo-collaborative portals to assist collaborative trip planning. (See the summaries of the four categories of online information in Table 1.)

2.4.2. Intra-group information sharing

In the proposed model (Fig. 3), for internal information acquisition, members of tourist groups share tourism information through different kinds of tools, either to make a point or ask for opinions. One participant said that, “I shared a link with them. It showed where I wanted to go so we could discuss it later.” With regard to the different types of tools for sharing information within a tourist group, Study 1 found that all the groups used communication tools, e.g., Facebook and WeChat (a popular application in China), and generally had positive views of the same. In family groups, participants preferred communicating face-to-face because this was the most efficient method. Participants in large groups (more than 5 people according to Quora, 2017) reported positive experiences in relation to using an online word processor that allowed real-time collaboration among users, e.g., Google Docs, which allows all group members to edit each other's comments, so that they could understand clearly each other's opinions. Participants also reported other approaches for communicating with each other, including making phone calls and sending emails.

With regard to the most commonly used tools (i.e., instant messaging applications, such as WhatsApp and WeChat), Study 1 found that people in large groups with less familiar group members required longer response times during the communication process. One participant in Study 1 said, “WeChat is good for initial discussions because I did not have to comment on someone's suggestion directly or immediately. I could just ignore the suggestion, so that person would know that the rest of us were not interested if no one responded and the discussions moved on to other topics.” Meanwhile, another participant identified one of the advantages of using instant messaging tools, “WeChat can save all the previous conversations so that I can read them later.”

2.4.3. Knowledge transfer

The “Infogineering Model” illustrates that knowledge is one of the foundations of actions. The proposed model illustrated in Fig. 3 shows that decisions are made based on members' knowledge, which consists of prior knowledge and knowledge from newly-processed information. On the one hand, prior knowledge is the knowledge that stems from previous experience. One participant said that, “I think I know Canada quite well, so I only searched information of certain places I heard of. It's (the holiday) only one week, I didn't

Table 1

Details of information acquisition found in Study 1.

Information acquisition	
Content	long distance transportation, local transportation, hotels, sightseeing attractions, activities, and food
Platform	search engines (e.g., Google), map services (e.g., Google Maps), travel websites (e.g., TripAdvisor), and social media (e.g., Facebook)
Format	pictures, videos, panoramas, maps, official textual introductions, travel strategy guides, review comments, blogs, and notifications from social media
Features	attraction popularity rankings, scores and reviews, recommendations based on interests, built-in maps, mash-up portals (i.e., integrated websites), links to original sources, and information based on geo-locations

even bother to search around.” On the other hand, tourists’ tourism knowledge is also gained from processing new information (from either internal or external information acquisition). It is a personalised interpretation process of decoding the received information, where different people may look at the same piece of information from different angles. Some people tend to ignore some aspects of the information, thus gaining a biased understanding, and leading to different decisions. Meanwhile, participants indicated that the variations in information interpretation through remote approaches (such as WhatsApp) could cause misunderstandings, so people tend to choose direct approaches (such as face-to-face) if possible. One participant explained that: *“He [her husband] was busy at work, so I did most of the planning, searched all sorts of information, and sent it to him for his opinion through either email or WeChat. But unfortunately [shot him a glare and smiled], he just browsed them quickly and never paid much attention. Every time, it was me who talked through everything later when he was home for dinner.”* This participant had not heard the other side of the story until her husband replied: *“That’s because it’s kind of tiring to read through what you sent me. I had to figure out what you wanted to say exactly. So why not wait until we meet and then talk?”* This finding is consistent with Alavi and Leidner’s (2001) work that knowledge is personalised, and for an individual’s knowledge to be useful to others, it must be expressed and communicated in a manner that is interpretable by the receivers. In mass-communications research, Hall (2001) describes the process of knowledge transfer among people through an encoding and decoding process.

Tourism knowledge is encoded into various types of information for transmission among group members (indicated by the broken line – Encoding, in Fig. 3). Received information from either an internal or external source is decoded and incorporated into individuals’ knowledge for decision making (indicated by the broken line – Decoding, in Fig. 3). Based on the EDT, knowledge *per se* cannot be transmitted, so information, as a medium, provides different forms to enable transmission. There are discrepancies in knowledge between the senders who encode the information, and the receivers who decode the same piece of information. Therefore, misunderstandings occur, which can lead to difficulties of communication, or even to disagreements and conflicts.

2.4.4. Decision making

Group holiday decisions are made based on the knowledge of all the active group members. Study 1 found that people used the same types of tools for sharing information, transferring knowledge, and making group decisions. The most popular type of tool is instant messaging, which provides significant support for the process of information sharing (discussed in Section 2.4.2). However, there are shortcomings when supporting knowledge transfer among group members, resulting in misunderstandings caused by different interpretations of the shared information (discussed in Section 2.4.3). At the final stage of the process, instant messaging applications support group decision making but with some weaknesses. For example, one participant commented that, *“When I missed a discussion session, I had to read through a lengthy exchange of discussions of other group members before arriving at their decisions.”* Similarly, one participant said, *“When some important decisions are made on WhatsApp, they will be displaced by latter messages. So people may miss them and ask again later. It is very annoying.”*

3. Study 2

3.1. Aim of study 2

While Study 1 found that technology plays an important role in the group holiday decision-making process, Study 2 further examined tourists’ preferences towards different types of online tourism information in terms of information acquisition and the use of tools for intra-group information sharing, knowledge transfer, and decision making. Furthermore, Study 2 explored how the use of technology is affected by cultural differences and group factors.

3.2. Related work

While many studies have explored the factors of the trip decision-making process, the majority are concerned with those of individual tourists, such as age (Peterson, 2007), gender (Kim, Lehto, & Morrison, 2007), nationality (Vuylsteke, Wen, Baesens, & Poelmans, 2010), motivation (Kozak, 2002), destination (Kim & Chalip, 2004), prior individual travel experience (Lehto, O’Leary, & Morrison, 2004), time, transportation, and cost constraints (Rugg, 1973), discretionary income (Engel & Blackwell, 1982), electronic word-of-mouth (Murphy, Mascardo, & Benckendorff, 2007), and destination image (Molina & Esteban, 2006).

Existing studies have also looked into the decision-making behaviour of tourist groups, such as the role of wives and husbands in joint family travel decisions (Mottiar & Quinn, 2004), the influences of children and their age in family holiday planning (Thornton et al., 1997), and family life cycles (Engel & Blackwell, 1982). Since the focus of this research is group tourists, Study 2 prioritised and focused mainly on group type characteristics. The novelty of Study 2 is that it explored a number of factors (related to the intrinsic characteristics of a tourist group) and investigated their influences on the group trip planning process supported by the use of technology. This section introduces the five factors examined in Study 2: cultural differences - nationality; tourist group factors - group type, prior group travel experience, group size, and group ties.

Nationality. Existing studies have found that nationality (cultural difference) influences people’s information searches, planning processes, and purchasing behaviour when making recreational trips (Gursoy & Chen, 2000). Gursoy and Umbreit (2004) identified different information search patterns among the citizens of EU countries because of their national cultural differences. Meanwhile, Vuylsteke et al. (2010) report that Chinese tourists prepare more prior to a long-distance trip and search more for relevant information when compared with Western tourists. Study 2 aimed to compare Chinese tourist groups and EU tourist groups in terms of their use of technology for group trip planning.

Group type. Existing literature has investigated the information search behaviour and trip preparation of different group types, where family groups have been studied from different aspects. The most popular topic is the distribution of work and roles among family members (Mottiar & Quinn, 2004). Meanwhile, the family lifecycle (including marital years, ages of children, retirement years, etc.) has been found to be influential on family holiday decision making (Collins & Tisdell, 2002). Study 2 also aimed to compare family groups and non-family groups in terms of their use of technology for group trip planning.

Group prior travel experience. The use of external information sources (the Internet, magazines, travel agencies, etc.) has been found to be different between first-time and frequent travellers (Chen & Gursoy, 2000). Lehto et al. (2004) indicated that tourists' activity anticipation and participation are influenced by their prior experience of a particular destination. Study 2 aimed to explore the extent of prior group travel experience with regard to the use of technology for group trip planning.

Group size. In their studies of group work, Levine and Moreland (1995) indicated that a group's size influences the nature of how group members interact in a number of ways. Smaller groups have stronger relationships and more intense dynamics, compared to larger groups. Members in larger groups rarely connect directly to all the other members and subgroups are very likely to form (Forsyth, 2009). Study 2 further aimed to examine whether group size plays a role in the use of technology for group trip planning.

Group ties. Another element in group work is the connection between group members (who-are-connected). Forsyth (2009) uses the term *group ties* to describe the connections and bonds that tie together the different members of a group. Group ties may be based on strong bonds, such as close family or friends, or weak bonds such as distant friends or strangers. Granovetter (1973) explained that weak (indirect) ties are easily broken with the passage of time or the occurrence of relationship-damaging events, and the nature of the ties influences group interaction. Finally, Study 2 also aimed to explore whether different levels of group ties (direct ties: close relatives, friends, colleagues; indirect ties: friends of friends, relatives of relatives, colleagues of colleagues, or strangers) influence tourists' use of technology for group trip planning.

3.3. Method

3.3.1. Study design

Study 2 employed a retrospective approach, questionnaire survey, asking participants to recall one of their previous trip experiences with a group of people. As shown in Fig. 4, the contribution of Study 2 is that it explores empirically how cultural differences (i.e., those of Chinese vs. Western tourists), and tourist group factors (i.e., group size, group ties, prior group travel experience, and group type) influence the use of technology for group trip planning from the perspectives of information acquisition, intra-group information sharing, knowledge transfer, and decision making.

To ensure the internal validity of this study, three variables were controlled to avoid interference. The study required each respondent to recall a past group travel experience that: (1) aimed for leisure or recreation (travel motivation), (2) a destination that they had not visited before (familiarity of the destination), (3) the group members were FITs (fully independent travellers) meaning that they had not travelled with tours or according to any imposed schedule.

3.3.2. Participants

Participants were recruited through posts on social media, and emails across the university campuses in both the UK and China. The only requirement for taking part in this survey, which was stated in the recruitment information sheet, was that participants had

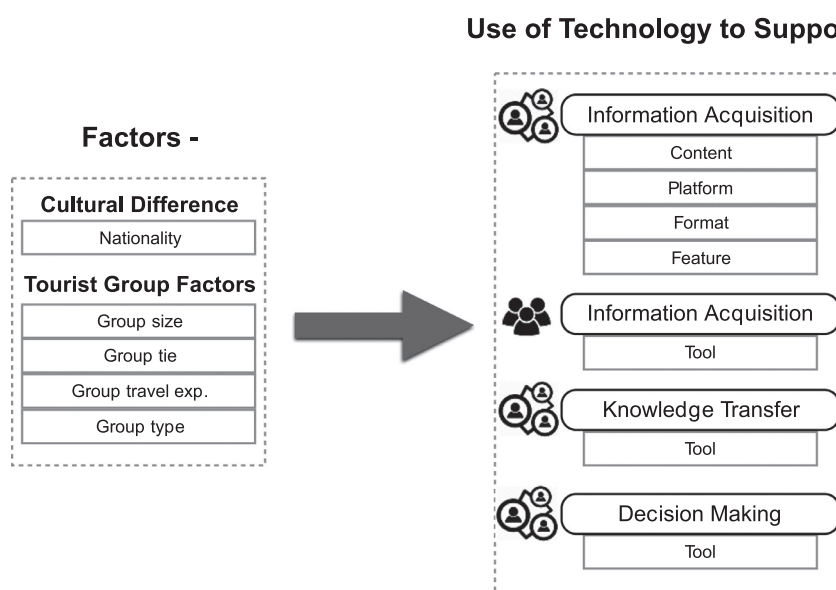


Fig. 4. Factors influencing the model for Study 2.

Table 2
Participants' background and analysis methods for Study 2.

Variables	Description	Data type	Method
Nationality	Chinese Westerners	Number = 198, Percentage = 75.9% Number = 37, Percentage = 24.1%	Nominal Chi-square
Group size	Pair (2 members) 3–5 members > 5 members	Number = 61, Percentage = 26% Number = 121, Percentage = 51.5% Number = 53, Percentage = 22.5%	Nominal Chi-square
Group tie	All direct ties At least one indirect tie	Number = 140, Percentage = 59.6% Number = 95, Percentage = 40.4%	Nominal Chi-square
Group type	Family Non-family	Number = 121, Percentage = 51.5% Number = 114, Percentage = 48.5%	Nominal Chi-square
GTE	Once per week Once per month Once per season Once per half-year Once per year	Number = 20, Percentage = 8.5% Number = 32, Percentage = 13.6% Number = 74, Percentage = 31.5% Number = 69, Percentage = 29.4% Number = 40, Percentage = 17.0%	5-point Likert (Ordinal) Chi-square
Gender	Male Female	Number = 88, Percentage = 37.4% Number = 147, Percentage = 62.6%	Nominal DA
Age	Mean = 29.312, SD = 12.207	Ratio	DA

GTE = group travel experience, DA = descriptive analysis.

to have had prior experience of travelling in a group within the preceding three months. In total, 235 responses were collected. See Table 2 for the participants' background information.

3.3.3. Materials

The questionnaire in Study 2 consisted of two parts. The first part included five questions. Four of the questions were closed-ended, obtaining the characteristics of the groups recalled by the participants, including nationality, group size (the number of people in the group in total), group ties (the number of direct friends, relatives, or colleagues in the group), and group type (a single-answer question from 'family', 'friends', 'colleagues', or 'mixed'). Then, a Likert scale question was used to measure the prior group travel experience of each participant.

The second part of the questionnaire asked the participants to recall their particular trip planning processes. Five multi-choice questions were posed to obtain information on their use of technology from two perspectives: information acquisition and the tools used for intra-group interaction. See the questions in Appendix C.

3.3.4. Data collection and analysis

The questionnaires were designed in English and then translated to Chinese for participants who could not read English. The online platforms used to collect the responses to the questionnaires were Qualtrics (English version) and Sojump (Chinese version). The participants completed the questionnaires individually and the average completion time was approximately eight minutes. No tangible reward was offered to the participants.

With regard to the data analysis, frequency and descriptive analysis were employed to gain an overall understanding of the participants. For the statistical analysis, a Chi-square test (Murphy et al., 2016) was employed to investigate any significant differences in the use of technology (nominal data) where group tourists exhibited different characteristics (i.e., nationality, group size, group ties, group type, and group travel experience). These characteristics were measured in terms of nominal and ordinal data. Although some of the sample sizes were significantly imbalanced (i.e., the Chinese sample size was 198, while the EU sample size was 37), the Chi-square tests make no assumptions about the equality of group sizes (McHugh, 2013). Meanwhile, the assumption for using Chi-square – '0 cells have an expected count less than 5' (Pallant, 2007) – was not violated in any of the analyses in this study. Existing studies (Fleming & Bowden, 2009; Jackson, 2001; Zeller, 2006) show that small sample sizes (i.e., between 10 and 50) are adequate for examination and assessment using factor analysis.

3.4. Results and discussion

3.4.1. Information acquisition

Information content. Study 2 shows that the most popular travel information for which people search is long distance transportation (chosen by 89.2% participants), followed by other popular information content: hotels (88.8%), sightseeing attractions (81.9%), local transportation (70.4%), food (64.2%), and activities (56.2%).

As shown in Table 3, the Chi-square indicates that: (1) Chinese tourists search for more information about sightseeing attractions (Chi-square = 8.68, $df = 1$, $p < 0.05$) and food-related information (Chi-square = 5.64, $df = 1$, $p < 0.05$), compared to tourists from the EU countries. Kivela and Crotts (2006) have found that food plays a major role in the way tourists experience a destination, and have found that travellers would return to the same destination to savour its unique gastronomy. This research empirically found that Chinese tourists have a higher priority for food whilst travelling. (2) Tourists in family groups tend to search for more information that is related to activities and events (Chi-square = 5.425, $df = 1$, $p < 0.05$), compared with tourists in non-family

Table 3
Influential factors and tourist groups' information content.

Factors	P values					
	Long distance transportation	Destination transportation	Hotels	Sightseeing attractions	Activities and events	Food, bars, restaurants
Nationality	0.061	0.355	0.130	0.004**	0.103	0.014*
GTE	0.176	0.563	0.008**	0.006**	0.457	0.481
Group type	0.352	0.357	0.264	0.192	0.020*	0.149

GTE = group travel experience.

*p value < 0.05, ** p value < 0.01.

groups. (3) Tourists with more group travel experience (who travel in groups once per month on average) search less for hotel information (Chi-square = 13.845, df = 4, $p < 0.05$) and pay less attention to sightseeing attraction information (Chi-square = 14.441, df = 4, $p < 0.05$), compared to tourists with less group travel experience. Existing literature focuses mainly on exploring how prior travel experience to a particular destination influences travellers' information search behaviour (Lehto et al., 2004) and attitudes towards such a destination (Lepp, 2008). This study has shifted the focus to groups and adds to the knowledge that prior group travel experience influences certain information searching behaviour.

Information platform. Study 2 shows that the most popular platform on which people search for tourism information is a search engine (chosen by 53.3% participants), while 43% of the participants preferred platforms for map and location services. Tourism websites, such as C-trip (the largest Chinese integrated tourism information website) and TripAdvisor, were chosen by 32.2% participants. Lastly, social media platforms, such as Facebook and Weibo (a popular Chinese social networking platform), were chosen by 14.8% participants.

As shown in Table 4, the Chi-square indicates that the tourists in family groups tend to use search engines for tourism information (Chi-square = 4.312, df = 1, $p < 0.05$), compared with tourists in non-family groups. With many studies examining the use of online websites and social media for tourism information (Xiang & Gretzel, 2010), this finding adds to the knowledge of how family groups use technology for tourism information.

Information format. Study 2 shows that the most popular travel information format that tourists choose to study pictures of the destination (chosen by 67.7% participants), followed by other popular information formats: travel strategy guides (66.2%), destination maps (61.9%), travel blogs (55.8%), textual reviews (51.2%), official textual travel information (37.7%), notifications and feeds from social media (27.3%), destination videos (24.2%), and destination panoramas (13%).

As shown in Table 5, the Chi-square indicates that: (1) Chinese tourists prefer viewing fewer destination pictures (Chi-square = 4.79, df = 1, $p < 0.05$), and (2) are also less likely to look at destination maps (Chi-square = 10.11, df = 1, $p < 0.05$), compared to people from the EU countries. This could be explained by information obtained in the interviews for Study 1 in which a Chinese participant explained that, "I usually rely on one or two members in our group to check the map and directions." (3) Tourists with less group travel experience (who travel with groups once per year on average) show a significant preference for reading textual reviews (Chi-square = 10.956, df = 4, $p < 0.05$). This finding could be explained by Ye, Law, Gu, and Chen (2011) who found that review comments have a significant influence on tourists' decision-making, and especially on decisions relating to accommodation (Gretzel & Yoo, 2008). (4) Tourists with relatively less group travel experience (who travel with groups once every six months) show significant interest in reading official textual introductions to destinations (Chi-square = 10.586, df = 4, $p < 0.05$). Official introductions, such as those which are government-sponsored, are considered to be credible and trustworthy (Cox et al., 2009).

Information feature. Study 2 shows that the most popular feature of travel information is the scores and comments from peer-travellers (chosen by 70% of participants), followed by other popular information features: recommendations based on personal interests (66.9%), destination attraction rankings (53.8%), mash-up travel information (48.5%), recommendations based on geo-location (30%), built-in maps (27.7%), and links to original sources (20%).

As shown in Table 6, the Chi-square indicates that: (1) Chinese tourists are less likely to check built-in maps (Chi-square = 4.63, df = 1, $p < 0.05$), but prefer considering more mash-up information websites (Chi-square = 10.35, df = 1, $p < 0.05$) – i.e., integrated tourism information portals with information from several aspects, including transportation, sightseeing, food, hotels, etc. Existing studies have explored the decision aid features of electronic commerce on consumers' decision performance (Tan, Teo, & Benbasat, 2010). Singh, Zhao, and Hu (2003) compares forty US-based international companies' domestic websites and their Chinese websites. Cultural differences give country-specific websites a look and feel that are unique to the local culture. To some extent, Study 2 reflects this finding because Chinese tourists, as consumers, prefer mash-up websites that have complicated front pages which

Table 4
Influential factors and tourist groups' information platform.

Factors	P values			
	Search engines	Map services	Tourism websites	Social media
Group type	0.048*	0.612	0.980	0.611

*p value < 0.05.

Table 5
Influential factors and tourist groups' information format.

Factors	p values								
	Picture	Video	Panorama	Map	Official textual info'	Travel guide	Textual review	Travel blog	Notification from social media
Nationality	0.030*	0.501	0.835	0.002**	0.455	0.542	0.772	0.467	0.194
GTE	0.489	0.426	0.575	0.678	0.032*	0.683	0.028*	0.478	0.457

GTE = group travel experience.

*p value < 0.05, **p value < 0.01.

Table 6
Influential factors and tourist groups' information features.

Factors	p values						
	Attraction popularity ranking	Scores, Review comments	Recommendation based on your interests	Built-in map	Mash-up portal	Links to original sources	Information based on geo-location
Nationality	0.104	0.352	0.756	0.026*	0.001**	0.249	0.089
Group type	0.010*	0.807	0.008**	0.746	0.391	0.172	0.607

*p value < 0.05, **p value < 0.01.

demonstrate the abilities to provide all aspects of tourism information. The Chi-square also indicates that tourists in family groups tend to check the rankings of popular sightseeing attractions at a destination (Chi-square = 6.679, df = 1, $p < 0.05$), and prefer recommendations based on personal interests (Chi-square = 6.943, df = 1, $p < 0.05$), compared with tourists in non-family groups. Among the relevant studies of family trip planning (Mottiar & Quinn, 2004; Thornton et al., 1997), this finding adds to the knowledge of what family tourist groups prefer to read when making their decisions.

3.4.2. Tools for information sharing, knowledge transfer and decision making

Study 2 shows that the most commonly used approach for sharing travel information, transferring knowledge, and making decisions within a tourist group is through instant messaging applications (chosen by 77.7% participants), such as WhatsApp and WeChat. The additional commonly used approaches include: phone calls (70.7%), face-to-face (66.2%), emails (11.5%), social media (such as Facebook and Weibo) (11.1%), and online word processors, such as Google Docs (4%), (although there was insufficient data for a statistical analysis).

As shown in Table 7, the Chi-square shows that Chinese tourist groups tend to use fewer emails to communicate among group members (Chi-square = 14.32, df = 1, $p < 0.05$), compared with tourists from other countries. The Chi-square also indicates that group size and group ties are found to be associated with how a group of tourists choose to share tourism information. (1) Groups with more than 5 members tend to use social media such as Facebook, compared to groups with fewer members (Chi-square = 7.16, df = 1, $p < 0.05$). (2) Groups having all direct ties (direct friends, relatives, and colleagues) prefer communicating face-to-face (Chi-square = 13.48, df = 1, $p < 0.05$), while groups having indirect ties (friends of friends, relatives of relatives, colleagues of colleagues, or strangers) prefer communicating through social media (Chi-square = 4.06, df = 1, $p < 0.05$). This result reflects the findings from Study 1, whereby people in larger groups with less familiar group members require longer response times during any intra-group communication.

4. Conclusion and implications

4.1. Conclusion

The tourism decision-making process for a tourist group requires effective communication for the sharing of relevant information (Decrop, 2005). This paper makes new contributions to the understanding of how information is obtained and shared within a tourist

Table 7
Influential factors and tourist groups' use of tools.

Factors	p values				
	Phone calls	Face-to-face	Emails	Instant message	Social media
Nationality	0.114	0.166	***	0.192	0.355
Group size	0.465	0.093	0.206	0.692	0.043*
Group tie	0.364	***	0.892	0.432	0.046*

*p value < 0.05, ***p value < 0.001.

group for the purposes of making decisions about their holiday. The model proposed in Study 1 is adapted from the IM and EDT in communication studies. It contains four components: information acquisition (internal and external information sources), intra-group information sharing (supported by different types of tools), knowledge transfer among group members through information (encoded by the senders and decoded by the receivers), and decision-making (based on the knowledge of active group members).

Furthermore, the findings of Study 2 add new features to the understanding of how the use of technology for group trip decision making is influenced by the intrinsic characteristics of tourist groups. Chinese tourists prefer to browse mash-up tourism portals for information, including background details, transportation, hotel, sightseeing attractions, etc. This could be explained by the previously defined cultural difference whereby East Asian cultures tend to focus on the context, and the relationships between object and the context (Nisbett & Miyamoto, 2005). Instant messaging applications, such as WhatsApp and WeChat, are commonly used by the majority of tourist groups. However, due to the need for privacy, people in larger groups or with weaker ties to each other tend to use social media applications, such as Facebook and Weibo, for intra-group communication. With regard to tourists with less prior group travel experience, they prefer accurate information (such as official introductions) about the essential parts of a trip (such as hotels). Finally, family groups tend to search directly through search engines, and have a need for the types of information that can deliver straightforward messages (such as attraction popularity rankings and lists of recommendations).

The findings of this research are useful and informative for researchers, who are in the field of tourism studies, for designers, who are interested in facilitating the group trip planning experience through designing new technical functions, interactions and interfaces, for online travel planners, who are interested in providing targeted readers with certain types of recommendations and information; and also for promoters of local tourism organisations, who may be interested in promoting the local tourism resources to different target groups.

The validity of the findings is also improved by the triangulation of using of the mixed-methods approach. By collecting different data types (i.e., qualitative and quantitative) generated from the two methods of group interviews and surveys, this research improved the validity of its findings. For example, the group holiday decision-making process and the use of technology were revealed in the group interviews, and were further validated in the questionnaire surveys.

4.2. Theoretical implications

Existing studies have proposed tourism decision-making models, which focus mainly on individual travellers (Decrop, 2005) and describe tourism in terms of space and time (Smallman & Moore, 2010). Study 1 proposes a model adapted from the 'Infogeering Model' and the EDT. It describes the process of how people in tourist groups make holiday decisions and illustrates the important role of technology in such a process. This model shifts the research focus from individual tourists to tourist groups, in that it describes the interactions between multiple users within a tourist group. It also highlights the role of technology in such a process and the interactions among group members. The model indicates that the decision-making process is not merely a form of communication between tourist and technology, but it also involves information sharing and processing that requires a higher level of cognitive effort for a group of travellers to make the most informed and desirable group decisions that will help to balance the interest of its members.

This model provides new insight, in the sense that it illustrates how the stages in the holiday decision-making process are connected with an underlying component of 'knowledge transfer' and an interactive process of 'encoding' and 'decoding'. On a surface level, tourism information is shared and passed around within a tourist group during the planning period. A question raised by the researchers was "what exactly is transferred through this action?". This model shows that it is *knowledge* that has been transferred, and decisions are made based on knowledge rather than information. This finding is well reflected in the "Infogeering Model" in which information is processed and converted to *knowledge* for decision making. *Knowledge* has been described as the ability to "drive people to action and decisions" (Bourdreau & Couillard, 1999). It is a cognitive process that group members read, comprehend, make sense of the received information, and convert it to *knowledge* that is exclusively characterised with personal interpretations. Existing studies have also reported some characteristics of group holiday decision making, such as conflicts, compromises, and delegations within a group (Decrop, 2005). They could be explained by this model in which a weak link lies in the 'encoding and decoding' process. Potentially, there are variations in terms of *knowledge* between the information senders and receivers, resulting in information loss and scenarios where senders and receivers may not understand each other fully. This can lead to conflicts, compromises, etc. Reflecting this finding in the EDT, information plays the role of a medium (meaningful discourse) to allow *knowledge* transmission among group members.

This model also provides insight, in the sense that it highlights the important role of technology in the group holiday decision-making process. Tourists in groups search through large amounts of tourism information on the Internet, and they search through different types of such information (i.e., different types of information content, format, platform, and feature). To achieve intra-group information sharing, knowledge transferring, and group decision making, digital tools (technologies) are used widely and the most commonly used type of tool is an instant messaging application, such as WhatsApp and WeChat. The majority of the comments about using such tools were positive because they support communication and interaction among group members. However, shortcomings and weaknesses were also reported in practice.

Existing studies have explored the various factors of a tourism decision-making process, of which the majority are concerned with the factors of individual tourists (e.g., age, gender, income, etc.) and destination (e.g., destination image, familiarity about the destination, etc.). Very few have explored the factors of tourist groups and the differences between different categories of tourist groups. This research has further developed a model that describes how the group decision-making process, when supported by technology, is influenced by factors related to the intrinsic characteristics of tourist groups. An enhanced and enriched model is presented (Fig. 4) which adds to the existing literature on tourism studies in terms of understanding the decision-making process of

group tourists. In the next section, more practical implications are provided for practitioners, namely designers, travel planners, and promoters of tourism organisations.

4.3. Practical implications

4.3.1. Facilitating information acquisition for different types of tourist groups

Firstly, the findings from Study 2 show that *Chinese tourists* have specific preferences when searching tourism information for a particular destination. The Chinese participants showed similarities to a holistic thinking pattern where they focused on multiple aspects of a destination when making a decision. It reflects the findings from Nisbett and Miyamoto (2005), in which members of East Asian cultures tend to engage in context-dependent and holistic perceptual processes when organising their environment. Consequently, this paper may provide suggestions for designers who target Chinese tourists – namely, to provide them with holistic tourism information for a destination. For example, creating a portal that integrates multiple aspects of tourism information, including flight tickets, rail tickets, car rentals, sightseeing attractions, hotels, travel agents, tourism guides, etc.

Secondly, Study 2 found that *family groups* have specific preferences when searching for tourism information. Prior studies on family decision making have shown that the Internet has become an important information source (Belch, Krentler, & Willis-Flurry, 2005). Meanwhile, since family holidays allow for family cohesiveness (Jiang, 2005) and self-development, especially for the experiential learning of children (Lehto, Fu, Li, & Zhou, 2017), tourism information searches by families are usually very purposeful. Therefore, this paper suggests that when targeting family tourist groups, providers should clearly present concise information about a sightseeing attraction or an activity, because families require a clear picture of the destinations to see if they fit the purposes of their particular trip. For example, providers could deliver information to families in highly summarised forms, such as attraction/activity rankings and lists of recommendations, so that they will know exactly what to expect.

Finally, Study 2 found that tourism information search preferences are influenced by prior group travel experience. Therefore, practitioners need to consider the different needs among people with different prior group travel experience, and provide different types of information accordingly. People with relatively more group travel experience (who travel once per month) prefer less information regarding hotels and sightseeing attractions, while those with the least amount of group travel experience (who travel only once per year or less) prefer more peer-generated information, such as review comments. People with relatively less group travel experience (who travel once every six months) require more textual official introductions about destinations, such as government-sponsored information.

4.3.2. Facilitating knowledge transfer among group members

Based on the proposed model for the decision-making process in group trip planning, misunderstandings can occur during the knowledge transfer process among group members, which reflects the findings from Hall (2001) in which the degree of “misunderstanding” in a communicative exchange depends on the degree of the non-identity of the codes of the receivers. The encoded information should be well structured and convey clear messages from the senders. Furthermore, the received information should be presented in line with the receivers’ thinking/meaning structures (Fuhse, 2009; Hall, 2001). Consequently, technology should help reduce any misunderstandings through supporting the encoding and decoding process at two levels: the presentation (user interface) and the significance (primary meaning) of information.

- On the level of the presentation of information, new types of interactions should be developed to simplify the encoding and decoding process, support the knowledge transfer, and increase the effectiveness of the information sharing among group members. For example, designers could consider providing suitable templates for senders to edit the information (such as sending an email message based on a template in Microsoft Outlook) and allowing receivers to choose their preferred interfaces to view the received information (such as setting a background and theme for an Internet browser on Chrome).
- On the level of the significance of information, new types of functions should be developed to highlight the significance of one piece of tourism information and further help the information receivers to assess its relevance. For example, technology can allow senders to manually tag a few key words of the information, so the receivers can easily decode the information. This is similar to, for example, the instructions on the Microsoft support websites, where bold font is used on certain keywords to better guide the reader. Furthermore, technology is able to implicitly uncover the underlying meanings of a piece of information, and key words could be labelled automatically in this way. Therefore, computational techniques could be developed in the holiday planning context, such as semantic analysis (Deerwester, Dumais, Furnas, Landauer, & Harshman, 1990) for textual tourism information, real time picture annotation techniques (Li & Wang, 2008) for tourism photographs, and annotation of streaming video techniques (Feng, Manmatha, & Lavrenko, 2004) for tourism videos.

4.3.3. Facilitating group decision making

This research has found that the most commonly used tools for group holiday decision making are instant messaging applications. In Study 1, participants in *large groups with indirect group ties* made specific comments about their group decision making using such applications. Firstly, although such instant messaging applications are not specifically designed for group trip planning, application designers should consider adding new features based on the original applications to facilitate group holiday decision making. For example, participants explained that it would be better if WhatsApp could allow users to set certain messages (such as very important messages) on top of the screen, so that all the group members could easily check any important messages at any time without having to ask repeatedly. Secondly, designers could allow built-in applications embedded in the original popular messaging applications to

make better use of their large customer base. Thus, users of the original applications will have easy access to this functionality for group trip planning whenever they become tourists. For example, “mini programs”, provided by WeChat (Yanes & Berger, 2017), allow small businesses to build their own commercial applications within the WeChat environment, as a result of which, the users of WeChat have been reported to find this very convenient to purchase products without switching to other applications (Wang, 2017).

All in all, communication within a group for group holiday decision making is only one type of communication using instant messaging applications. Therefore, to facilitate group holiday decision making through technology, designers should consider co-operating with popular social network companies, and develop new features and applications based on their existing environments.

5. Limitations

This research contributes to the knowledge of group trip planning and sheds light on implications for both theory and practice. However, there are a number of limitations: (1) the methods employed for Study 2 were based on retrospective recalls of participants' previous travelling experience. Participants were asked to recall a group trip within the preceding three months, but this may still have caused recall bias; (2) the controlled variables in designing Study 2 only covered limited aspects - travel motivation, familiarity with the destination, and travel type, and we acknowledge that there may be other variables such as travel period, destination image, destination distance, language of the destination, etc. These variables could influence the internal validity of this study. This study kept these factors open to recruit as many participants with recent group travelling experience as possible; (3) the imbalanced level of data in Study 2 (i.e., between the Chinese and the EU participants, and between female and male participants) may have influenced the validity of the results to some extent.

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Appendix A. Structure and questions of Study 1

Structure	Questions
Infogineering Model	How did you get the holiday idea? How did you get the holiday-related information? What information did you search for? Think of an example and describe the process by which you obtained information. How did you make decisions within your group? Think of an example and describe the decision making process.
Theory of encoding and decoding	Did any disagreement or conflict occur during the decision making? If yes, please describe how it happened and how it was resolved. Did you use online services to help you get tourism information? What services? What about the services that you liked most? e.g., functionalities, format Did you use online services to help you communicate with your group? What services? What about the services that you like most? e.g., functionalities, platforms Do you have any comments on the online services you used to communicate with your group members?

Appendix B. Participant profile in Study 1

Group type	Size	Gender	Age range	Nationality	Use of online media in planning
1. Family	2	Female	30–39	British	5 – Highly depends on it
		Male	30–39	British	5 – Highly depends on it
	2 Minor	Female	< 10	British	N/A
2. Family	2	Female	40–49	British	4 – Always
		Male	40–49	British	3 – Once in a while
	1 Minor	Male	10–16	British	N/A
3. Family	3	Male	30–39	British	4 – Always
		Female	30–39	British	N/A
		Female	30–39	Indonesian	4 – Always

	1 Minor	Male	16–18	British	N/A
4. Friend	2	Female	18–29	Chinese	5 – Highly depends on it
		Female	18–29	Chinese	3 – Once in a while
5. Friend	3	Female	18–29	Greek	4 – Always
		Female	18–29	Greek	4 – Always
		Female	18–29	Greek	5 – Highly depends on it
6. Friend	2	Male	30–39	Chinese	5 – Highly depends on it
		Male	30–39	Chinese	5 – Highly depends on it
7. Colleague	3	Male	18–29	Chinese	5 – Highly depends on it
		Female	30–39	Ghanaian	4 – Always
		Female	50–59	British	3 – Once in a while
8. Colleague	3	Male	30–39	Canadian	4 – Always
		Female	30–39	Canadian	N/A
		Female	30–39	Chinese	4 – Always
9. Colleague	2	Female	30–39	Chinese	5 – Highly depends on it
		Male	30–39	Chinese	N/A
10. Mixed	3	Male	18–29	Chinese	5 – Highly depends on it
		Female	18–29	Chinese	5 – Highly depends on it
		Male	18–29	Chinese	5 – Highly depends on it
11. Mixed	> 3	Female	18–29	Chinese	3 – Always
12. Mixed	> 3	Male	30–39	Chinese	3 – Always

The people highlighted in this table did not take part in the study, but their involvements in the trip planning process were described by group members who took part in the study.

Appendix C

Multiple choice questions of group tourists' use of technology.

Question: *What information _____ will you prefer/use/search during your group trip planning process?*

Multiple choices

Content	long distance transportation, destination transportation, hotels, sightseeing attractions, activities and events, food
Platform	Search engines (Google, Baidu), Tourism websites (Tripadvisor, C-trip) Map (Google maps, Baidu maps), Social media (Facebook)
Format	destination pictures, destination videos, destination panoramas, destination maps, textual official travel information, travel strategy guides, textual reviews, other travellers' travel stories, notifications from social media
Feature	attraction rankings on popularity, scores and reviews from other users, information based on your interests, built-in maps, mash-up travel information, links to original sources, information based on geo-locations
Tool	phone calls, face-to-face, emails, Instant messaging Apps (WhatsApp, WeChat), Social media (Facebook, Twitter), online word processor (Google Docs)

References

- Alavi, M., & Leidner, D. E. (2001). Review: Knowledge management and knowledge management systems: Conceptual foundations and research issues. *MIS Quarterly*, 25(1), 107–136.
- Belch, M. A., Krentler, K. A., & Willis-Flurry, L. A. (2005). Teen internet mavens: Influence in family decision making. *Journal of Business Research*, 58(5), 569–575.
- Blair, E. (2015). A reflexive exploration of two qualitative data coding techniques. *Journal of Methods and Measurement in the Social Sciences*, 6(1), 14–29.
- Bourdreau, A., & Couillard, G. (1999). Systems integration and knowledge management. *Information Systems Management*, 16, 24–32.
- Bronner, F., & De Hoog, R. (2011). Vacationers and eWOM: Who posts, and why, where, and what? *Journal of Travel Research*, 50(1), 15–26.
- Carr, N. (2006). A comparison of adolescents' and parents' holiday motivations and desires. *Tourism and Hospitality Research*, 6(2), 129–142.
- Chen, J. S., & Gursoy, D. (2000). Cross-cultural comparison of the information sources used by first-time and repeat travelers and its marketing implications. *International Journal of Hospitality Management*, 19(2), 191–203.
- Chen, Y. C., Shang, R. A., & Li, M. J. (2014). The effects of perceived relevance of travel blogs' content on the behavioral intention to visit a tourist destination. *Computers in Human Behavior*, 30, 787–799.
- Collins, D., & Tisdell, C. (2002). Gender and differences in travel life cycles. *Journal of Travel Research*, 41(2), 133–143.
- Cox, C., Burgess, S., Sellitto, C., & Buultjens, J. (2009). The role of user-generated content in tourists' travel planning behaviour. *Journal of Hospitality Marketing & Management*, 18(8), 743–764.
- Decrop, A. (2005). Group processes in vacation decision-making. *Journal of Travel & Tourism Marketing*, 18(3), 23–36.
- Decrop, A., & Snelders, D. (2004). Planning the summer vacation: An adaptable process. *Annals of Tourism Research*, 31(4), 1008–1030.
- Deerwester, S., Dumais, S. T., Furnas, G. W., Landauer, T. K., & Harshman, R. (1990). Indexing by latent semantic analysis. *Journal of the American Society for Information Science*, 41(6), 391.
- Dipietro, R. B., Wang, Y., Rompf, P., & Severt, D. (2007). At-destination visitor information search and venue decision strategies. *International Journal of Tourism Research*, 9(3), 175–188.

- Engel, J., & Blackwell, R. (1982). *Consumer behavior (fourth edition)*. Hinsdale: The Dryden Press.
- Feng, S. L., Mamatha, R., & Lavrenko, V. (2004). Multiple bernoulli relevance models for image and video annotation. *Proceedings of the 2004 IEEE computer society conference*. 2 II–II.
- Fleming, C., & Bowden, M. (2009). The most commonly cited disadvantages of web-based surveys are sample frame and non-response bias. *Journal of Environmental Management*, 90(1), 284–292.
- Fodness, D. (1992). The impact of family life cycle on the vacation decision-making process. *Journal of Travel Research*, 31, Fall, 8–13.
- Forsyth, D. (2009). *Group Dynamics* (5th edition). Belmont: Wadsworth.
- FreshMinds Research. The role of social media in holiday destination decision making, prepared for VisitBritain, (2011)., Accessed 22.09.16, https://www.visitbritain.org/sites/default/files/vb-corporate/Documents-Library/documents/Social_media_in_the_holiday_decision_making_process_v3.pdf.
- Fuhse, J. A. (2009). The meaning structure of social networks. *Sociological Theory*, 27(1), 51–73.
- Gnoth, J. (1997). Tourism motivation and expectation formation. *Annals of Tourism Research*, 24(2), 283–304.
- Goodall, B. (1991). *Understanding holiday choice. Progress in Tourism. Recreation and hospitality management*, 3, London: Belhaven Press 59–77.
- Granovetter, M. S. (1973). The strength of weak ties. *American Journal of Sociology*, 78, 1360–1380.
- Gretzel, U., & Yoo, K. H. (2008). Use and impact of online travel reviews. *Information and Communication Technologies in Tourism 2008*, 35–46.
- Gursoy, D., & Chen, J. S. (2000). Competitive analysis of cross cultural information search behaviour. *Tourism Management*, 21(6), 583–590.
- Gursoy, D., & Umbreit, W. T. (2004). Tourist information search behavior: Cross-cultural comparison of European Union member states. *International Journal of Hospitality Management*, 23(1), 55–70.
- Hall, S. (2001). Encoding/decoding. Media and Cultural Studies: Keywords. In M. G. Durham, & D. M. Kellner (Eds.). Blackwell Publishing.
- Infogineering Model. Accessed 15th July (2017)., <http://www.infogineering.net/infogineering-model.htm>.
- Jackson, D. L. (2001). Sample size and number of parameter estimates in maximum likelihood confirmatory factor analysis: A Monte Carlo investigation. *Structural Equation Modeling*, 8(2), 205–223.
- Jafari, J. (1982). The tourism market basket of goods and services: The components and nature of tourism. *The Tourism Market Basket of Goods and Services: The Components and Nature of Tourism*, 1–12.
- Jiang, H. (2005). Rational thinking on family entertainment style. *Theory of Practice and Education*, (9), 14–16.
- Jupp, V. (Ed.). (2006). *The sage dictionary of social research methods*. Sage.
- Kaplan, A. M., & Haenlein, M. (2010). Users of the world, unite! The challenges and opportunities of social media. *Business Horizons*, 53(1), 59–68.
- Kim, N., & Chalip, L. (2004). Why travel to the FIFA world cup? Effects of motives, background, interest, and constraints. *Tourism Management*, 25(6), 695–707.
- Kim, D., Lehto, X. Y., & Morrison, A. M. (2007). Gender differences in online travel information search: Implications for marketing communications on the internet. *Tourism Management*, 28(2), 423–433.
- Kivela, J., & Crofts, J. C. (2006). Tourism and gastronomy: Gastronomy's influence on how tourists experience a destination. *Journal of Hospitality & Tourism Research*, 30(3), 354–377.
- Kozak, M. (2002). Comparative analysis of tourist motivations by nationality and destinations. *Tourism Management*, 23(3), 221–232.
- Kvale, S., & Brinkmann, S. (2009). *Interviews: Learning the craft of qualitative research interviewing*. Los Angeles: Sage Publication.
- Lehto, X. Y., Fu, X., Li, H., & Zhou, L. (2017). Vacation benefits and activities: Understanding Chinese family travellers. *Journal of Hospitality & Tourism Research*, 41(3), 301–328.
- Lehto, X. Y., O'Leary, J. T., & Morrison, A. M. (2004). The effect of prior experience on vacation behaviour. *Annals of Tourism Research*, 31(4), 801–818.
- Lepp, A. (2008). Attitudes towards initial tourism development in a community with no prior tourism experience: The case of Bigodi, Uganda. *Journal of Sustainable Tourism*, 16(1), 5–22.
- Leung, D., Law, R., van Hoof, H., & Buhalis, D. (2013). Social media in tourism and hospitality: A literature review. *Journal of Travel & Tourism Marketing*, 30(1–2), 3–22.
- Levine, J. M., & Moreland, R. L. (1995). *Group processes. Advanced social psychology*. New York: McGraw-Hill 419–465.
- Lew, A., & McKercher, B. (2006). Modeling tourist movements: A local destination analysis. *Annals of Tourism Research*, 33(2), 403–423.
- Li, J., & Wang, J. Z. (2008). Real-time computerized annotation of pictures. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 30(6), 985–1002.
- Litvin, S. W., Goldsmith, R. E., & Pan, B. (2008). Electronic word-of-mouth in hospitality and tourism management. *Tourism Management*, 29(3), 458–468.
- Lo, I. S., Mckercher, B., Lo, A., & Cheung, C. S. (2011). Tourism and online photography. *Tourism Management*, 32(4), 725–731.
- Mansfeld, Y. (1993). *Tourism: Towards a behavioral approach*. Oxford: Pergamon Press.
- Marlow, J., & Dabbish, L. (2014). When is a picture not worth a thousand words? The psychological effects of mediated exposure to a remote location. *Computers in Human Behaviour*, 30, 824–831.
- Mathieson, A., & Wall, G. (1982). *Tourism, economic, physical and social impacts*. Longman.
- McHugh, M. L. (2013). The chi-square test of independence. *Biochemia Medica*, 23(2), 143–149.
- McNamara, C. (2009). “General guidelines for conducting interviews,” Accessed 05.07.17, <http://managementhelp.org/businessresearch/interviews.html>.
- Middleton, V. T. C., & Clarke, J. R. (2001). *Marketing in travel and tourism* (Third ed.). London: Butterworth-Heinemann.
- Molina, A., & Esteban, A. (2006). Tourism brochures: Usefulness and image. *Annals of Tourism Research*, 33(4), 1036–1056.
- Mottiar, Z., & Quinn, D. (2004). Couple dynamics in household tourism decision-making: Women as the gatekeepers? *Journal of Vacation Marketing*, 10, 149–160.
- Murphy, L., Mascardo, G., & Benckendorff, P. (2007). Exploring word-of-mouth influences on travel decisions: Friends and relatives vs. other travellers. *International Journal of Consumer Studies*, 31(5), 517–527.
- Murphy, H. C., Chen, M. M., & Cossutta, M. (2016). An investigation of multiple devices and information sources used in the hotel booking process. *Tourism Management*, (52), 44–51.
- Nisbett, R. E., & Miyamoto, Y. (2005). The influence of culture: Holistic versus analytic perception. *Trends in Cognitive Sciences*, 9(10), 467–473.
- Quora. “What does GIT mean in the tourism industry?” Accessed 24th July (2017)., <https://www.quora.com/What-does-GIT-mean-in-the-tourism-industry>.
- Pallant, J. (2007). *SPSS survival manual: A step-by-step guide to data analysis using SPSS version 15*. Nova Iorque: McGraw Hill.
- Peterson, M. (2007). Effects of income, assets and age on the vacationing behavior of US consumers. *Journal of Vacation Marketing*, 13(1), 29–43.
- Prebensen, N. K. (2005). Segmenting the group tourist heading for warmer weather. *Journal of Travel & Tourism Marketing*, 19(4), 27–40.
- Rugg, D. (1973). The choice of journey destination: A theoretical and empirical analysis. *Review of Economics and Statistics*, 55(1), 64–72.
- Ross, E. L. D., & Iso-Ahola, S. E. (1991). Sightseeing tourists' motivation and satisfaction. *Annals of tourism research*, 18(2), 226–237.
- Sandelowski, M. (2000). Focus on research methods-whatever happened to qualitative description? *Research in Nursing and Health*, 23(4), 334–340.
- Schmoll, G. (1977). *Tourism promotion*. London: Tourism International Press.
- Seybold, P. B. (2001). Get inside the lives of your customer's. *Harvard Business Review*, 79(5), 80–89 May.
- Sigala, M. (2010). Measuring customer value in online collaborative trip planning processes. *Marketing Intelligence & Planning*, 28(4), 418–443.
- Singh, N., Zhao, H., & Hu, X. (2003). Cultural adaptation on the web: A study of American companies' domestic and Chinese websites. *Journal of Global Information Management (JGIM)*, 11(3), 63–80.
- Smallman, C., & Moore, K. (2010). Process studies of tourists' decision-making. *Annals of Tourism Research*, 37(2), 397–422.
- Sun, X., Sharples, S., & Makri, S. (2011). A user-centred mobile diary study approach to understanding serendipity in information research. *Information Research*, 16(3), 492–493.
- Tan, C. H., Teo, H. H., & Benbasat, I. (2010). Assessing screening and evaluation decision support systems: A resource-matching approach. *Information Systems Research*, 21(2), 305–326.
- Thornton, P. R., Shaw, G., & Williams, A. M. (1997). Tourist group holiday decision-making and behaviour: The influence of children. *Tourism Management*, 18(5), 287–297.
- Vuylsteke, A., Wen, Z., Baesens, B., & Poelmans, J. (2010). Consumers' Search for Information on the Internet: How and why China differs from Western Europe.

- Journal of Interactive Marketing*, 24(4), 309–331.
- Wang, Y. (2017). "How Chinese super app WeChat plans to lock out foreign app stores in China," Accessed 03.08.17, <https://www.forbes.com/sites/ywang/2017/01/09/chinese-super-app-wechat-launches-new-plan-to-rival-app-stores-in-china/#7da3da165748>.
- Wang, K. C., Hsieh, A. T., Yeh, Y. C., & Tsai, C. W. (2004). Who is the decision-maker: The parents or the child in group package tours? *Tourism Management*, 25(2), 183–194.
- Wong, B. L. W., & Blandford, A. E. (2002). Analysing ambulance dispatcher decision making: Trailing emergent themes analysis. *Proceedings of the joint conference of the computer human interaction special interest group and the ergonomics society of Australia HF2002*, Publish: Factors 2002.
- World Bank (2017). "International tourism, expenditures," Accessed 20th Jul, <http://data.worldbank.org/indicator/ST.INT.XPND.CD>.
- Xiang, Z., & Gretzel, U. (2010). Role of social media in online travel information search. *Tourism Management*, 31(2), 179–188.
- Yanes, P. A., & Berger, P. D. (2017). How WeChat has changed the face of marketing in China. *British Journal of Marketing Studies*, 5(3), 14–21.
- Ye, Q., Law, R., Gu, B., & Chen, W. (2011). The influence of user-generated content on traveler behavior: An empirical investigation on the effects of e-word-of-mouth to hotel online bookings. *Computers in Human Behavior*, 27(2), 634–639.
- Zeller, R. A. (2006). "Statistical tools in applied research," Accessed 05.04.17, <http://www.personal.kent.edu/rzeller/Ch.%2010.pdf>.